敏實科技大學
MINTH UNIVERSITY OF SCIENCE AND TECHNOLOGY
人工智慧專業大學

MITUST · Nguyên lý xe tự lái · Tuần 6

Ba cảm biến không dùng để nhìn: siêu âm, GNSS, IMU

三種不是用來看的感測器：超音波、衛星定位、 IMU

Siêu âm chỉ thấy hai mét · vệ tinh nói bạn ở đâu · quán tính nói bạn đã quay bao nhiêu

超音波只看兩公尺· 衛星告訴你在哪· IMU 告訴你轉了多少

工具｜ **Công cụ** ：
ch-neural.github.io/MITUST-Principles-of-Self-Driving-Cars/week6/spec/

第六週 第 2 節　2026/10/15（四）10:10–11:00　CH Tseng

Phần 1 / 3

Siêu âm: rẻ nhất, gần nhất, chậm nhất

超音波：最便宜、最近、最慢

Hai datasheet: một mô-đun vài chục nghìn, một linh kiện ô tô

兩份規格表：一顆幾十塊的模組、一顆車用的

Siêu âm · 兩份規格表

Ví dụ · sáu người cùng gọi trong thung lũng: ai gọi tên mình mới nhận ra tiếng vọng
例子· 六個人在山谷同時喊，各喊自己的名字才認得出回音

Hai cảm biến siêu âm đo xa gần như nhau; bản ô tô biết tự phát hiện bị che

兩顆超音波測距差不多，車用那顆會自己發現被遮住

Mô-đun học tập · HC-SR04
教學模組 · HC-SR04

Electric Parameter

Working Voltage	DC 5 V
Working Current	15mA
Working Frequency	40Hz
Max Range	4m
Min Range	2cm
MeasuringAngle	15 degree
Trigger Input Signal	10uS TTL pulse
Echo Output Signal	Input TTL lever signal and the range in proportion
Dimension	45*20*15mm

ElecFreaks, Ultrasonic Ranging Module HC-SR04, p.1 · 節錄 · 讀取 2026/10/08

Cảm biến ô tô · Bosch USS Gen 6
車用 · Bosch 超音波第六代

Key features	Ultrasonic sensors generation		
	6.0	6.1	6.5
Object presence detection	6 cm	6 cm	3 cm
Min. measuring range[3]	15 cm	15 cm	15 cm
Max. measuring range[4]	2,5 m	4,5 m	5,5 m
ASIL sensor/ECU	QM/QM	A/B	B²/B
Blindness detection	Yes	Yes	Yes
Adaptive threshold	Yes	Yes	Yes
Signal coding	Yes	Yes	Yes
Digital sig. processing	Yes	Yes	Yes
Fusional capability (video/radar etc.)	Yes	Yes	Yes

[3] Reference object: 75mm standard pole
[4] Measures & process in accordance with ASIL B, FIT<10-7/h

Bosch Aftermarket, Ultrasonic Sensors Generation 6 factsheet (03/2023), p.1 · 節錄 · 讀取 2026/10/08

Video: HC-SR04 đo khoảng cách thế nào · 影片：HC-SR04 怎麼量距離 2:02 youtu.be/vf2lW4LkmMQ

4 m và 5,5 m gần như nhau; khác ở chỗ biết mình bị bẩn che, và không nhầm tiếng vọng của cái khác.
4 公尺對 5.5 公尺差不多；差在會不會自己發現被髒污蓋住、會不會跟別顆的回波搞混。

Vật lý · 為什麼超音波只能看幾公尺

Ví dụ · thấy chớp trước, vài giây sau mới nghe sấm
例子· 先看到閃電，幾秒後才聽到雷聲



Tốc độ âm 340 m/s: 5,5 m đi về mất 32 ms — trên 20 km/h là không kịp

聲速 340 m/s ：5.5 公尺來回 32 毫秒——時速 20 以上就來不及

20 km/h cần quãng phanh 4,2 m cộng dự phòng 5 m, vượt quá 4–5,5 m nó thấy được. Nên siêu âm chỉ làm một việc: đỗ xe, lùi xe, bù điểm mù ở tốc độ thấp.

時速 20 煞停要 4.2 公尺加餘裕 5 公尺，超過它看得到的 4 到 5.5 公尺。所以超音波只做一件事：停車、倒車、低速補盲。

Video: cảm biến lùi xe qua ba đời (tiếng Trung) · 影片：倒車雷達的進化史 3:18 youtu.be/lV-MiiOEGIA

«Bãi đỗ xe» (5 km/h, 2 m) là chỗ duy nhất hôm nay nó qua được.
「停車場」（時速 5、2 公尺）是今天唯一讓它過關的地方。

Phần 2 / 3

GNSS: cho vị trí tuyệt đối, nhưng không phải lúc nào cũng có

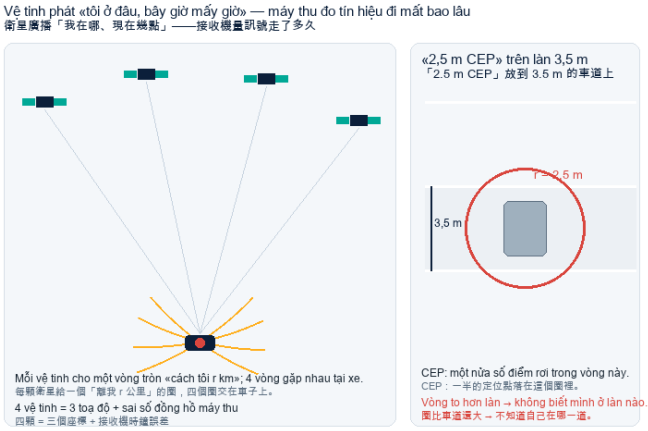
衛星定位 GNSS：給你絕對位置，但不是隨時都有

2,5 m hay 1 cm, khác nhau ở một dòng và một trạm gốc

2.5 公尺還是 1 公分，差在一行字和一座基準站

Nguyên lý · 衛星定位的原理

Ví dụ · ba mốc cho biết «bạn cách tôi bao xa» — giao điểm là bạn
例子· 三個地標各說「你離我多遠」，圈的交點就是你



示意圖：課程自製（ src/tools_make_diagrams.py ）

Video: GPS định vị thế nào (tiếng Trung) 4:41
影片：GPS 定位原理（中文動畫） youtu.be/R_mY4GLhauk

Bốn vệ tinh, ba toạ độ cộng một đồng hồ: máy thu đơn lẻ sai 2 đến 10 m

四顆衛星、三個座標加一個時鐘：單機精度 2 到 10 公尺

Mỗi vệ tinh phát «tôi ở đâu, bây giờ mấy giờ»; máy thu đo tín hiệu đi mất bao lâu, nhân tốc độ ánh sáng là khoảng cách. Một làn đường chỉ rộng 3,5 m.
每顆衛星廣播自己在哪、現在幾點；接收機量訊號走了多久，乘光速就是距離。一個車道才 3.5 公尺寬。

GPS chỉ là một hệ: GPS, GLONASS, Galileo, Bắc Đẩu gộp lại gọi là GNSS.
GPS 只是其中一家：GPS、GLONASS、Galileo、北斗合起來叫 GNSS。

1 / 2 u-blox NEO-M8N（導航等級）

Ví dụ · ném 100 phi tiêu, một nửa rơi trong vòng
例子· 射一百支飛鏢，一半落在圈裡

GNSS		GPS & GLONASS	GPS	GLONASS	BeiDou	Galileo
Horizontal position accuracy [3]	With SBAS	2.5 m 2.0 m	2.5 m 2.0 m	4 m -	3 m -	3 m -
NEO-M8N/Q						
Max navigation update rate	NEO-M8N	5 Hz	10 Hz	10 Hz	10 Hz	10 Hz
	NEO-M8Q	10 Hz	18 Hz	18 Hz	18 Hz	18 Hz
Time-To-First-Fix [4]	Cold start	26 s	29 s	30 s	34 s	45 s
	Hot start	1 s	1 s	1 s	1 s	1 s
	Aided starts [5]	2 s	2 s	2 s	3 s	7 s

[3] CEP, 50%, 24 hours static, -130 dBm, > 6 SVs

u-blox NEO-M8 Data sheet (UBX-15031086 R14), §1.3 Table 1 與註腳 3（中間省略）· 節錄 · 讀取 2026/10/08



NEO-M8 圖：規格書封面

NEO-M8N: 2,5 m CEP — một nửa số điểm nằm trong vòng

NEO-M8N ： 2.5 m CEP ，一半的定位點在圈內

Độ chính xác
精度

2,5 m; có SBAS 2,0 m
2.5 m ；有 SBAS 修正 2.0 m

Chú thích 3
註腳 3

CEP 50% · đứng yên 24 h · −130 dBm · > 6 vệ tinh
CEP 50%· 靜止 24 小時· − 130 dBm· 6 顆以上衛星

Cập nhật · khởi động
更新· 開機

5 Hz; khởi động lạnh 26 s
5 Hz ；冷開機 26 秒

Vòng 2,5 m to hơn làn đường: nó biết bạn ở đường nào, không biết ở làn nào.
2.5 公尺的圈比車道還大：它知道你在哪條路，不知道你在哪一道。

2 / 2 u-blox ZED-F9P（公分級）

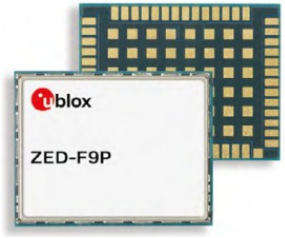
Ví dụ · đồng hồ lớp nhanh 5 phút, có người cầm đồng hồ đúng báo cho cả lớp
例子· 教室的鐘快 5 分鐘，有人拿準的錶告訴大家

GNSS		GPS+GLO+GAL+BDS	GPS+GLO+GAL	GPS+GAL	GPS+GLO	GPS+BDS	GPS
Horizontal pos. accuracy	PVT[7]	1.5 m CEP	1.5 m CEP	1.5 m CEP	1.5 m CEP	1.5 m CEP	1.5 m CEP
	SBAS[7]	1.0 m CEP	1.0 m CEP	1.0 m CEP	1.0 m CEP	1.0 m CEP	1.0 m CEP
	RTK[8]	0.01 m + 1 ppm CEP	0.01 m + 1 ppm CEP	0.01 m + 1 ppm CEP	0.01 m + 1 ppm CEP	0.01 m + 1 ppm CEP	0.01 m + 1 ppm CEP
Vertical pos. accuracy	RTK[8]	0.01 m + 1 ppm R50	0.01 m + 1 ppm R50	0.01 m + 1 ppm R50	0.01 m + 1 ppm R50	0.01 m + 1 ppm R50	0.01 m + 1 ppm R50

Table 3: ZED-F9P-02B position accuracy in different GNSS modes

u-blox ZED-F9P-02B Data sheet (UBX-21023276 R03), §1.2 Table 3 · 節錄 · 讀取 2026/10/08

Video: u-blox giải thích RTK 2:34
影片：u-blox 講 RTK youtu.be/28_YpdN8Thw



ZED-F9P 圖：規格書封面

Độ chính xác RTK của ZED-F9P: 1 cm, cộng 1 mm cho mỗi km cách trạm gốc

ZED-F9P 的 RTK 精度：1 公分，加上離基準站每公里 1 公釐

RTK
RTK

0,01 m + 1 ppm CEP (trạm gốc ≤ 20 km)
0.01 m + 1 ppm CEP（基準站 20 km 內）

Không hiệu chỉnh
沒修正時

PVT 1,5 m CEP: mất hiệu chỉnh là lùi về cấp mét
PVT 1.5 m CEP ：修正一斷就退回公尺級

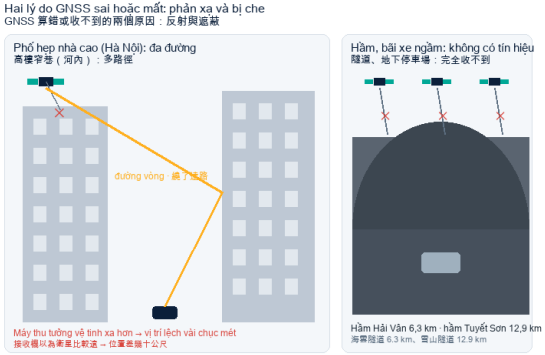
Hội tụ
收斂

< 10 s; cập nhật 8 Hz (bốn hệ)
< 10 s ；更新 8 Hz（四系統）

RTK cần ai đó gửi hiệu chỉnh (e-GNSS, VNGEONET). Không có trạm gốc, nó là máy thu 1,5 m.
RTK 要有人送修正（e-GNSS、VNGEONET）。沒有基準站，它就是一顆 1.5 公尺的接收機。

Thất bại · 衛星定位什麼時候失靈

Ví dụ · nhìn người trong gương, tưởng họ đứng sau gương
例子· 看鏡子裡的人，會以為他在鏡子後面



示意圖：課程自製（ src/tools_make_diagrams.py ）

- **Video: GPS là gì (tiếng Trung) 1:45**
  影片：什麼是 GPS youtu.be/u54tYw5_NsY
- **Video: vào hầm là mất GPS 0:52**
  影片：進隧道就沒 GPS youtu.be/U6rOEcyo6xU

Hai chỗ GNSS hỏng: phố hẹp nhà cao (phản xạ) và đường hầm (bị che)

衛星定位失靈的兩個地方：高樓窄巷（反射）、隧道（遮蔽）

Ngõ hẹp, nhà cao: tín hiệu dội tường rồi mới vào, đi đường dài hơn, máy thu tưởng vệ tinh xa hơn — lệch vài chục mét mà không biết.
Trong hầm: không nhận được gì.
窄巷高樓：訊號從牆上反射再進來，走了遠路，接收機以為衛星比較遠——偏幾十公尺還不自知。隧道：什麼都收不到。

Trên bàn đọc: ZED-F9P qua ở sân trường, mất khoá ở Hà Nội lùi về 1,5 m, trong hầm bằng không.
判讀台上： ZED-F9P 校園過、河內失鎖退回 1.5 m 、隧道零。

Phần 3 / 3

Quán tính IMU: không cần bên ngoài — nên nó trôi

IMU ：不靠外界，所以會飄

Hai datasheet: chip trong điện thoại và mô-đun công nghiệp

兩份規格表：手機裡那顆和工業級那顆

1 / 2 TDK ICM-20948（手機等級）

Ví dụ · không ai đứng lên, cân vẫn chỉ 2 kg
例子· 沒人站上去，體重計也顯示 2 公斤

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
GYROSCOPE SENSITIVITY						
Full-Scale Range	GYRO_FS_SEL=0		±250		dps	1
	GYRO_FS_SEL=1		±500		dps	1
	GYRO_FS_SEL=2		±1000		dps	1
	GYRO_FS_SEL=3		±2000		dps	1
Gyroscope ADC Word Length			16		bits	1
Sensitivity Scale Factor	GYRO_FS_SEL=0		131		LSB/(dps)	1
	GYRO_FS_SEL=1		65.5		LSB/(dps)	1
	GYRO_FS_SEL=2		32.8		LSB/(dps)	1
	GYRO_FS_SEL=3		16.4		LSB/(dps)	1
Sensitivity Scale Factor Tolerance	25°C		±1.5		%	2
Sensitivity Scale Factor Variation Over Temperature	-40°C to +85°C		±3		%	2
Nonlinearity	Best fit straight line; 25°C		±0.1		%	2, 3
Cross-Axis Sensitivity			±2		%	2, 3
ZERO-RATE OUTPUT (ZRO)						
Initial ZRO Tolerance	25°C (Component-level)		±5		dps	2
ZRO Variation Over Temperature	-40°C to +85°C		±0.05		dps/°C	2
GYROSCOPE NOISE PERFORMANCE (GYRO_FS_SEL=0)						
Noise Spectral Density	Based on Noise Bandwidth = 10 Hz		0.015		dps/√Hz	2
GYROSCOPE MECHANICAL FREQUENCIES		25	27	29	kHz	2
LOW PASS FILTER RESPONSE	Programmable Range	5.7		197	Hz	1, 3
GYROSCOPE START-UP TIME	From Full-Chip Sleep mode		35		ms	2, 3
OUTPUT DATA RATE	Low-Power Mode	4.4		562.5	Hz	1
	Low-Noise Mode GYRO_FCHOICE=1; GYRO_DLPFCFG=x	4.4		1.125k	Hz	
	Low-Noise Mode GYRO_FCHOICE=0; GYRO_DLPFCFG=x			9k	Hz	

Table 1. Gyroscope Specifications

TDK InvenSense, ICM-20948 Data Sheet (DS-000189 v1.3), §3.1 Table 1 · 節錄 · 讀取 2026/10/08

Video: bên trong con quay MEMS (Bosch) 1:30
影片：Bosch 車用陀螺儀裡面 youtu.be/XsjvaYAFN1M

ICM-20948 (điện thoại): đứng yên vẫn đọc 5 °/s, không ghi ổn định lệch không

ICM-20948（手機級）：靜止也讀到 5 °/s ，沒有偏差穩定度那一行

Lệch không
初始零偏

Initial ZRO ±5 dps: đứng yên vẫn đọc ra, phải tự hiệu chuẩn

Initial ZRO ±5 dps ：靜止也會讀到，要自己校

Nhiệt độ
溫度

±0,05 dps/°C: chênh 10 °C là 0,5 °/s

±0.05 dps/℃ ：差 10 ℃ 就是 0.5 °/s

Dòng không có
沒有的那行

In-run bias stability: datasheet này không có

In-run bias stability ：這份規格書沒有

Ứng dụng datasheet tự ghi: điện thoại, máy tính bảng, thiết bị đeo. Không ghi xe.
規格書自己寫的用途：手機、平板、穿戴。沒寫車。

2 / 2 Analog Devices ADIS16470（工業級）

Ví dụ · đồng hồ rẻ lệch 1 phút/ngày, đồng hồ tốt vài giây/tháng
例子· 便宜的錶一天差一分鐘，好錶一個月才差幾秒

Parameter	Test Conditions/Comments	Min	Typ	Max	Unit
GYROSCOPES					
Dynamic Range		±2000			°/sec
Sensitivity	16-bit format		10		LSB/°/sec
	32-bit format		655,360		LSB/°/sec
Error over Temperature	−10°C ≤ T_C ≤ +75°C		±0.25		%
Misalignment	Axis to axis		±0.1		Degrees
Nonlinearity[1]	Full scale (FS) = 2000°/sec		±0.25		%FS
Bias					
In Run Stability	1 σ		8		°/hr
Angular Random Walk	1 σ		0.34		°/√hr
Error over Temperature	−10°C ≤ T_C ≤ +75°C, 1 σ		0.2		°/sec
Linear Acceleration Effect	Any direction, 1 σ		0.015		°/sec/*g*
Vibration Rectification Error			0.0005		°/sec/g^2
Output Noise	1 σ, no filtering		0.17		°/sec rms
Rate Noise Density	1 σ, f = 10 Hz to 40 Hz		0.008		°/sec/√Hz rms
3 dB Bandwidth			550		Hz
Sensor Resonant Frequency			66		kHz
ACCELEROMETERS[2]	Each axis				

Analog Devices, ADIS16470 Data Sheet Rev. C, p.3 Table 1 · 節錄 · 讀取 2026/10/08



ADIS16470 裝在評估板 ADIS16470/PCBZ 上 圖： Analog Devices

ADIS16470 (công nghiệp): ổn định lệch không 8 °/h, ghi ngay dưới mục Bias

ADIS16470（工業級）：偏差穩定度 8 °/h ，寫在 Bias 底下

Ổn định lệch không
偏差穩定度

Bias → In Run Stability 8 °/hr (1 σ)

Bias ⟶ In Run Stability 8 °/hr （ 1 σ ）

Random walk
隨機游走

Angular Random Walk 0,34 °/√hr

Angular Random Walk 0.34 °/√hr

Dải đo
量程

Dynamic Range ±2000 °/sec — như cấp điện thoại

Dynamic Range ±2000 °/sec—— 和手機級一樣

Trang 1 tự ghi ứng dụng: «Unmanned and autonomous vehicles». Chip điện thoại ghi: smartphones.
第 1 頁自己寫的用途：「無人與自駕載具」。手機級那顆寫的是：手機。

Hai cấp IMU khác ở độ ổn định lệch không (200 lần), không ở dải đo

兩級 IMU 差在偏差穩定度（兩百多倍），不是量程

Dòng trên datasheet 規格表上那一行	ICM-20948 (điện thoại) ICM-20948（手機級）	ADIS16470 (công nghiệp) ADIS16470（工業級）	Nghĩa là gì 這代表什麼
Ổn định lệch không 偏差穩定度	không in (lớp ước 1800 °/h) 沒印（本課估 1800 °/h）	8 °/h	Khác hơn 200 lần 差兩百多倍
Angular random walk 角度隨機游走	không in 沒印	0.34 °/√h	Nhiễu tích luỹ theo √t 雜訊累積，開根號時間
Dải đo 量程	±2000 °/s	±2000 °/s	Giống nhau — không phải khác ở đây 一樣——不是差在這裡
Giá 價格	vài USD 幾塊美元	vài trăm USD 幾百美元	Khác 100 lần 差一百倍

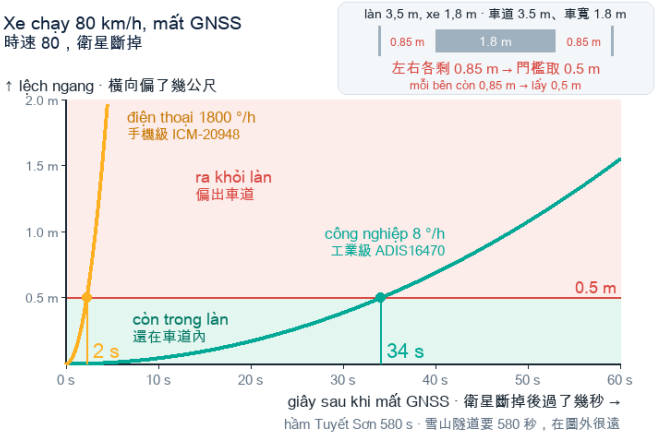
Trên nữa còn cấp sợi quang: giữ vài phút, vài chục nghìn USD. Cấp = giữ được bao lâu.

再上去還有光纖等級：撐幾分鐘、幾萬美元。等級就是撐多久。

Ví dụ · hai chiếc đồng hồ đều có mặt 12 giờ; khác ở chỗ chạy đúng hay không
例子· 兩支錶的錶面都是 12 小時，差在走得準不準

Tính thử · 撐多久

Ví dụ · nhắm mắt đi thẳng: vài bước đầu không sao, càng đi càng lệch
例子· 閉著眼睛走直線：前幾步沒事，越走越偏



示意圖：課程自製（ src/tools_make_diagrams.py ）

IMU giữ vài chục giây: lệch ngang ≈ v × trôi × t² ÷ 2

IMU 撐幾十秒：橫向誤差 ≈ 車速 × 偏差 × 時間² ÷ 2

80 km/h, cho phép lệch 0,5 m: cấp công nghiệp 8 °/h giữ được 34 s, cấp điện thoại 2 s. Hầm Tuyết Sơn 12,9 km cần 580 s — không cái nào giữ nổi.

時速 80、容許 0.5 公尺：8 °/h 的工業級撐 34 秒，手機級撐 2 秒。雪山隧道 12.9 公里要撐 580 秒——沒有一顆撐得住。

IMU bù lỗ vài giây đến vài chục giây, không phải để qua hầm. Qua hầm nhờ thứ khác (chương 5).
IMU 補幾秒到幾十秒的洞，不是用來過隧道。過隧道要靠別的（第五章）。

Nhìn lại · OS1 裡面的 IMU

Ví dụ · vừa đi vừa chụp ảnh toàn cảnh, ảnh ghép bị méo
例子· 邊走邊拍全景照，接起來的照片會歪

Trong OS1 có sẵn một IMU: 640 mẫu mỗi giây

OS1 裡面就有一顆 IMU ：每秒 640 筆

Lidar quay một vòng mất 100 ms, trong 100 ms đó xe đi 1,1 m, point cloud bị kéo méo. IMU trong máy là để «kéo thẳng» vòng đó. Datasheet còn ghi rõ: TDK IAM-20680HT.
光達轉一圈要 100 毫秒，車子在這 100 毫秒裡走了 1.1 公尺，點雲會被拉歪。機內的 IMU 就是用來把這一圈的點「拉直」的。規格書還指名是 TDK 的 IAM-20680HT 。

IMU OUTPUT	
Connection	UDP over 1000Base-T or 1000Base-T1
Samples Per Second	640
Data Per Sample	3 axis gyro, 3 axis accelerometer
Timestamp Resolution	< 1 µs
Data Latency	< 10 ms
Additional Details	InvenSense IAM-20680HT; datasheet for more details: https://invensense.tdk.com/download-pdf/iam-20680ht-datasheet/

Ouster OS1 datasheet, Rev7 / FW 3.2, p.3 · 節錄 · 讀取 2026/10/08

«IMU pose correction matrix» tuần trước chính là tham số ngoài giữa nó và thân lidar.
上週第二節講的「IMU pose correction matrix」就是它和光達本體的外參。

Thử nhanh · 小練習（ 4 分鐘動手 ）

Mở công cụ nhỏ số ③: kéo độ trôi từ 8 lên 1800, rồi kéo tốc độ từ 80 xuống 5.

打開判讀台第③個小工具：把偏差從 8 拉到 1800 ，再把車速從 80 拉到 5 。

1. **«Giữ được bao lâu» từ 34 s thành mấy giây?**
   「撐多久」從 34 秒變幾秒？

2. **Bãi đỗ xe 5 km/h: cấp điện thoại có giữ nổi 60 s không?**
   時速 5 的停車場，手機級撐得過 60 秒嗎？

3. **Tình huống nào ngay cả 8 °/h cũng không giữ nổi?**
   哪一個情境連 8 °/h 都撐不住？

Đáp án nằm trong câu 4 bài tập tiết sau; giờ chỉ cần cảm nhận đường cong đó.
答案在下一節的作業第四題；現在先感覺一下那條曲線。

Tiết 2 · 第二節小結

Ba câu của tiết 2

第二節的三句話

Siêu âm: 5,5 m, trên 20 km/h là vô dụng, nhưng ở bãi đỗ xe nó là vua — rẻ nhất và duy nhất thấy được 20 cm trước cản.

超音波：5.5 公尺、時速 20 以上沒用，但停車場它是王——最便宜、唯一看得到保險桿前 20 公分。

GNSS: 2,5 m là xác suất (CEP 50%); RTK 1 cm phải có người gửi hiệu chỉnh; nhà cao thì lệch, trong hầm thì mất.

GNSS ： 2.5 m 是機率（ CEP 50% ）； RTK 1 公分要有人送修正；高樓會偏，隧道會斷。

IMU: không cần bên ngoài nên trôi; cấp = giữ được bao lâu (8 °/h: 34 s ở 80 km/h). Nó bù lỗ vài giây, không đi qua hầm.

IMU ：不靠外界所以會飄；等級就是撐多久（ 8 °/h 在時速 80 撐 34 秒）。它補幾秒的洞，不過隧道。

Phụ lục · 附錄（不計時）

HC-SR04 — toàn bộ tài liệu gốc (2 trang)

HC-SR04 — 文件原頁（全兩頁）

ELEC Freaks Tech Support: services@elecfreaks.com

Ultrasonic Ranging Module HC - SR04

Product features:

Ultrasonic ranging module HC - SR04 provides 2cm - 400cm non-contact measurement function, the ranging accuracy can reach to 3mm. The modules includes ultrasonic transmitters, receiver and control circuit. The basic principle of work:
(1) Using IO trigger for at least 10us high level signal,
(2) The Module automatically sends eight 40 kHz and detect whether there is a pulse signal back.
(3) IF the signal back, through high level , time of high output IO duration is the time from sending ultrasonic to returning.
Test distance = (high level time×velocity of sound (340M/S) / 2,

Wire connecting direct as following:

- 5V Supply
- Trigger Pulse Input
- Echo Pulse Output
- 0V Ground

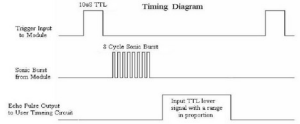
Electric Parameter

Working Voltage	**DC 5 V**
Working Current	**15mA**
Working Frequency	**40Hz**
Max Range	**4m**
Min Range	**2cm**
MeasuringAngle	**15 degree**
Trigger Input Signal	**10uS TTL pulse**
Echo Output Signal	**Input TTL lever signal and the range in proportion**
Dimension	**45*20*15mm**



Timing diagram

The Timing diagram is shown below. You only need to supply a short 10uS pulse to the trigger input to start the ranging, and then the module will send out an 8 cycle burst of ultrasound at 40 kHz and raise its echo. The Echo is a distance object that is pulse width and the range in proportion .You can calculate the range through the time interval between sending trigger signal and receiving echo signal. Formula: uS / 58 = centimeters or uS / 148 =inch; or: the range = high level time * velocity (340M/S) / 2; we suggest to use over 60ms measurement cycle, in order to prevent trigger signal to the echo signal.



ElecFreaks, Ultrasonic Ranging Module HC-SR04 · 原頁全文 · 讀取 2026/10/08 · 課堂引用

Phụ lục · 附錄（不計時）

Bosch Ultrasonic Sensors Generation 6 — nguyên trang factsheet

Bosch 超音波第六代 — 規格單原頁



Invented for life

BOSCH

Bosch Ultrasonic Sensors Generation 6

Reliable distance measurement for safe and comfortable parking and maneuvering

Bosch ultrasonic sensors (USS) detect distances to obstacles and monitor space when parking and maneuvering. The USS generation 6 is a nonimpact distance sensor that works on the basis of ultrasound for automotive applications. It uses an echo technology with a variable ultrasonic frequency in the range of 43kHz – 60kHz.

Key features	Ultrasonic sensors generation		
	6.0	6.1	6.5
Object presence detection	6 cm	6 cm	3 cm
Min. measuring range[2]	15 cm	15 cm	15 cm
Max. measuring range[2]	2,5 m	4,5 m	5,5 m
ASIL sensor/ECU	QM/QM	A/B	B[1]/B
Blindness detection	Yes	Yes	Yes
Adaptive threshold	Yes	Yes	Yes
Signal coding	Yes	Yes	Yes
Digital sig. processing	Yes	Yes	Yes
Fusional capability (video/radar etc.)	Yes	Yes	Yes

[1] Reference object: 75mm standard pole
[2] Measures & process in accordance with ASIL B, FIT<10-7/h

Advantages at a glance:

- One sensor covers multiple brands
- Easy installation: plug and play
- Sensor for aftermarket is based on Bosch OE sensor
- Basis for parking and maneuvering systems as well as for braking and automated parking functions
- Enables handling in very small parking lots (vehicle + 80 cm)
- Sensitive ultrasonic system (e.g. detection of low reflecting objects)
- Very robust against dirt, ice and challenging environmental conditions

AA-TR/PAE, 03/2023, Content and images subject to technical and program changes.

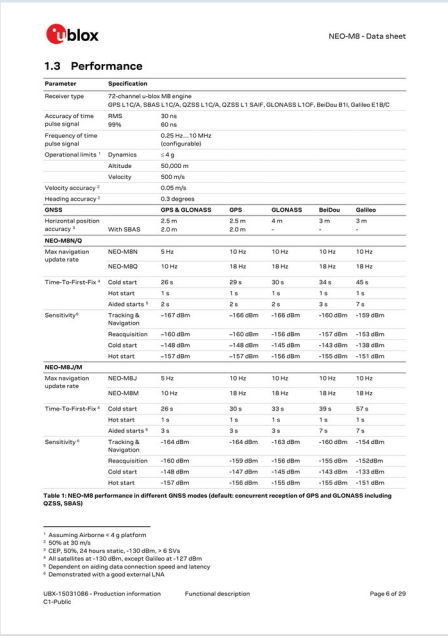
boschaftermarket.com

Bosch Aftermarket, Bosch Ultrasonic Sensors Generation 6 factsheet (AA-TR/PAE, 03/2023), p.1 · 原頁全文 · 讀取 2026/10/08 · 版權歸 Bosch ，課堂引用

Phụ lục · 附錄（不計時）

u-blox NEO-M8 (trang 6) và ZED-F9P-02B (trang 4–5) — nguyên trang

u-blox NEO-M8（第 6 頁）與 ZED-F9P-02B（第 4–5 頁）— 原頁



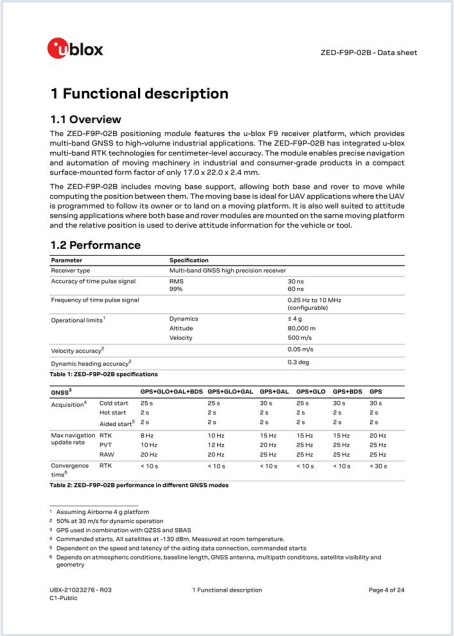
NEO-M8 - Data sheet

1.3 Performance

Parameter		Specification				
Receiver type		72-channel u-blox M8 engine GPS L1C/A, SBAS L1C/A, QZSS L1C/A, QZSS L1 SAIF, GLONASS L1OF, BeiDou B1I, Galileo E1B/C				
Accuracy of time pulse signal	RMS 99%	30 ns 60 ns				
Frequency of time pulse signal		0.25 Hz...10 MHz (configurable)				
Operational limits [1]	Dynamics	≤ 4 g				
	Altitude	50,000 m				
	Velocity	500 m/s				
Velocity accuracy [2]		0.05 m/s				
Heading accuracy [2]		0.3 degrees				
GNSS		**GPS & GLONASS**	**GPS**	**GLONASS**	**BeiDou**	**Galileo**
Horizontal position accuracy [3]	With SBAS	2.5 m 2.0 m	2.5 m 2.0 m	4 m -	3 m -	3 m -
NEO-M8N/Q						
Max navigation update rate	NEO-M8N	5 Hz	10 Hz	10 Hz	10 Hz	10 Hz
	NEO-M8Q	10 Hz	18 Hz	18 Hz	18 Hz	18 Hz
Time-To-First-Fix [4]	Cold start	26 s	29 s	30 s	34 s	45 s
	Hot start	1 s	1 s	1 s	1 s	1 s
	Aided starts [5]	2 s	2 s	2 s	3 s	7 s
Sensitivity [6]	Tracking & Navigation	−167 dBm	−166 dBm	−166 dBm	−160 dBm	−159 dBm
	Reacquisition	−160 dBm	−160 dBm	−156 dBm	−157 dBm	−153 dBm
	Cold start	−148 dBm	−148 dBm	−145 dBm	−143 dBm	−138 dBm
	Hot start	−157 dBm	−157 dBm	−156 dBm	−155 dBm	−151 dBm
NEO-M8J/M						
Max navigation update rate	NEO-M8J	5 Hz	10 Hz	10 Hz	10 Hz	10 Hz
	NEO-M8M	10 Hz	18 Hz	18 Hz	18 Hz	18 Hz
Time-To-First-Fix [4]	Cold start	26 s	30 s	33 s	39 s	57 s
	Hot start	1 s	1 s	1 s	1 s	1 s
	Aided starts [5]	3 s	3 s	3 s	7 s	7 s
Sensitivity [6]	Tracking & Navigation	−164 dBm	−164 dBm	−163 dBm	−160 dBm	−154 dBm
	Reacquisition	−160 dBm	−159 dBm	−156 dBm	−155 dBm	−152dBm
	Cold start	−148 dBm	−147 dBm	−145 dBm	−143 dBm	−133 dBm
	Hot start	−157 dBm	−156 dBm	−155 dBm	−155 dBm	−151 dBm

Table 1: NEO-M8 performance in different GNSS modes (default: concurrent reception of GPS and GLONASS including QZSS, SBAS)

[1] Assuming Airborne < 4 g platform
[2] 50% at 30 m/s
[3] CEP, 50%, 24 hours static, -130 dBm, > 6 SVs
[4] All satellites at -130 dBm, except Galileo at -127 dBm
[5] Dependent on aiding data connection speed and latency
[6] Demonstrated with a good external LNA



ZED-F9P-02B - Data sheet

1 Functional description

1.1 Overview

The ZED-F9P-02B positioning module features the u-blox F9 receiver platform, which provides multi-band GNSS to high-volume industrial applications. The ZED-F9P-02B has integrated u-blox multi-band RTK technologies for centimeter-level accuracy. The module enables precise navigation and automation of moving machinery in industrial and consumer-grade products in a compact surface-mounted form factor of only 17.0 x 22.0 x 2.4 mm.

The ZED-F9P-02B includes moving base support, allowing both base and rover to move while computing the position between them. The moving base is ideal for UAV applications where the UAV is programmed to follow its owner or to land on a moving platform. It is also well suited to attitude sensing applications where both base and rover modules are mounted on the same moving platform and the relative position is used to derive attitude information for the vehicle or tool.

1.2 Performance

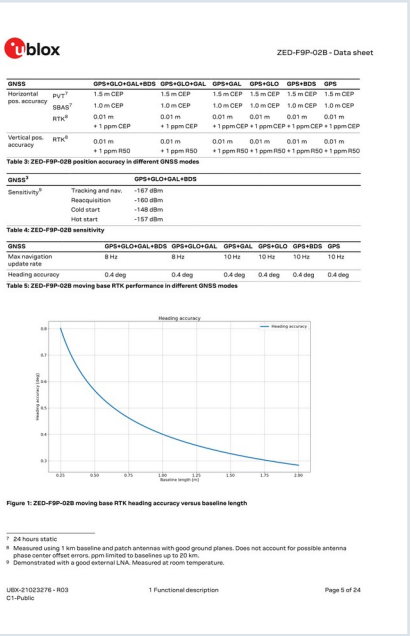
Parameter	Specification	
Receiver type	Multi-band GNSS high precision receiver	
Accuracy of time pulse signal	RMS 99%	30 ns 60 ns
Frequency of time pulse signal		0.25 Hz to 10 MHz (configurable)
Operational limits [1]	Dynamics Altitude Velocity	≤ 4 g 80,000 m 500 m/s
Velocity accuracy [2]		0.05 m/s
Dynamic heading accuracy [2]		0.3 deg

Table 1: ZED-F9P-02B specifications

GNSS [9]		GPS+GLO+GAL+BDS	GPS+GLO+GAL	GPS+GAL	GPS+GLO	GPS+BDS	GPS
Acquisition [4]	Cold start	25 s	25 s	30 s	25 s	30 s	30 s
	Hot start	2 s	2 s	2 s	2 s	2 s	2 s
	Aided start [5]	2 s	2 s	2 s	2 s	2 s	2 s
Max navigation update rate	RTK	8 Hz	10 Hz	15 Hz	15 Hz	15 Hz	20 Hz
	PVT	10 Hz	12 Hz	20 Hz	25 Hz	25 Hz	25 Hz
	RAW	20 Hz	20 Hz	25 Hz	25 Hz	25 Hz	25 Hz
Convergence time [8]	RTK	< 10 s	< 10 s	< 10 s	< 10 s	< 10 s	< 30 s

Table 2: ZED-F9P-02B performance in different GNSS modes

[1] Assuming Airborne 4 g platform
[2] 50% at 30 m/s for dynamic operation
[3] GPS used in combination with QZSS and SBAS
[4] Commanded starts. All satellites at -130 dBm. Measured at room temperature.
[5] Dependent on the speed and latency of the aiding data connection, commanded starts
[6] Depends on atmospheric conditions, baseline length, GNSS antenna, multipath conditions, satellite visibility and geometry



ZED-F9P-02B - Data sheet

GNSS		GPS+GLO+GAL+BDS	GPS+GLO+GAL	GPS+GAL	GPS+GLO	GPS+BDS	GPS
Horizontal pos. accuracy	PVT [7]	1.5 m CEP	1.5 m CEP	1.5 m CEP	1.5 m CEP	1.5 m CEP	1.5 m CEP
	SBAS [7]	1.0 m CEP	1.0 m CEP	1.0 m CEP	1.0 m CEP	1.0 m CEP	1.0 m CEP
	RTK [8]	0.01 m + 1 ppm CEP	0.01 m + 1 ppm CEP	0.01 m + 1 ppm CEP	0.01 m + 1 ppm CEP	0.01 m + 1 ppm CEP	0.01 m + 1 ppm CEP
Vertical pos. accuracy	RTK [8]	0.01 m + 1 ppm R50	0.01 m + 1 ppm R50	0.01 m + 1 ppm R50	0.01 m + 1 ppm R50	0.01 m + 1 ppm R50	0.01 m + 1 ppm R50

Table 3: ZED-F9P-02B position accuracy in different GNSS modes

GNSS [9]		GPS+GLO+GAL+BDS
Sensitivity [8]	Tracking and nav.	−167 dBm
	Reacquisition	−160 dBm
	Cold start	−148 dBm
	Hot start	−157 dBm

Table 4: ZED-F9P-02B sensitivity

GNSS	GPS+GLO+GAL+BDS	GPS+GLO+GAL	GPS+GAL	GPS+GLO	GPS+BDS	GPS
Max navigation update rate	8 Hz	8 Hz	10 Hz	10 Hz	10 Hz	10 Hz
Heading accuracy	0.4 deg	0.4 deg	0.4 deg	0.4 deg	0.4 deg	0.4 deg

Table 5: ZED-F9P-02B moving base RTK performance in different GNSS modes

Figure 1: ZED-F9P-02B moving base RTK heading accuracy versus baseline length

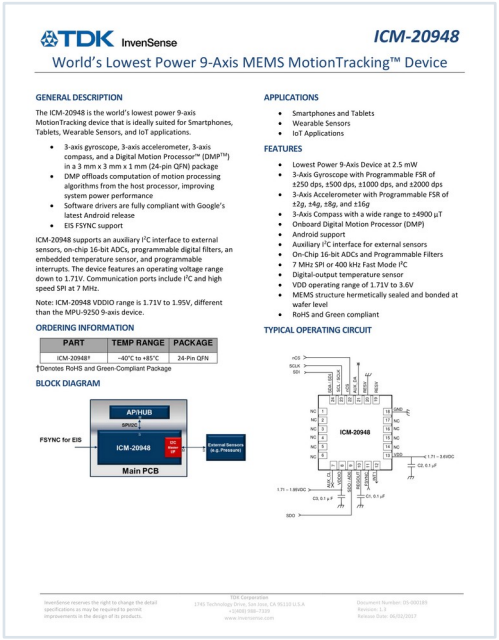
[7] 24 hours static
[8] Measured using 1 km baseline and patch antennas with good ground planes. Does not account for possible antenna phase center offset errors. ppm limited to baselines up to 20 km.
[9] Demonstrated with a good external LNA. Measured at room temperature.

u-blox NEO-M8 Data sheet UBX-15031086 R14 p.6 ； ZED-F9P-02B Data sheet UBX-21023276 R03 p.4–5 · 原頁全文 · 讀取 2026/10/08 · 版權歸 u-blox ，課堂引用

Phụ lục · 附錄（不計時）

TDK ICM-20948 — trang bìa và trang 11 (thông số con quay)

TDK ICM-20948 — 封面與第 11 頁（陀螺儀規格）



TDK InvenSense

ICM-20948

World's Lowest Power 9-Axis MEMS MotionTracking™ Device

GENERAL DESCRIPTION

The ICM-20948 is the world's lowest power 9-axis MotionTracking device that is ideally suited for Smartphones, Tablets, Wearable Sensors, and IoT applications.

- 3-axis gyroscope, 3-axis accelerometer, 3-axis compass, and a Digital Motion Processor™ (DMP™) in a 3 mm x 3 mm x 1 mm (24-pin QFN) package
- DMP offloads computation of motion processing algorithms from the host processor, improving system power performance
- Software drivers are fully compliant with Google's latest Android release
- EIS FSYNC support

ICM-20948 supports an auxiliary I^2C interface to external sensors, on-chip 16-bit ADCs, programmable digital filters, an embedded temperature sensor, and programmable interrupts. The device features an operating voltage range down to 1.71V. Communication ports include I^2C and high speed SPI at 7 MHz.

Note: ICM-20948 VDDIO range is 1.71V to 1.95V, different than the MPU-9250 9-axis device.

ORDERING INFORMATION

PART	TEMP RANGE	PACKAGE
ICM-20948†	-40°C to +85°C	24-Pin QFN

†Denotes RoHS and Green-Compliant Package

BLOCK DIAGRAM

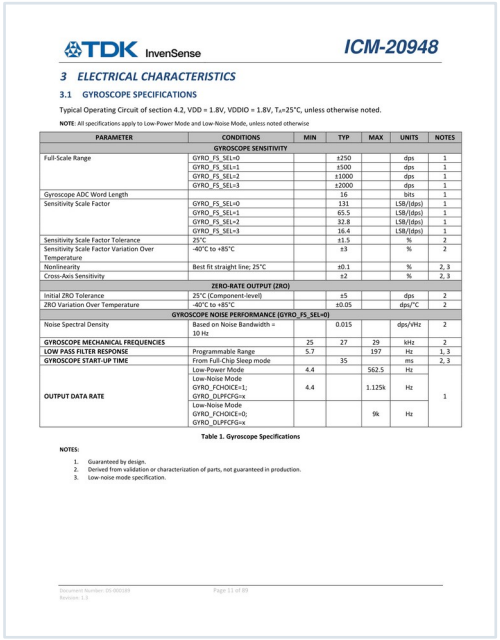
APPLICATIONS

- Smartphones and Tablets
- Wearable Sensors
- IoT Applications

FEATURES

- Lowest Power 9-Axis Device at 2.5 mW
- 3-Axis Gyroscope with Programmable FSR of ±250 dps, ±500 dps, ±1000 dps, and ±2000 dps
- 3-Axis Accelerometer with Programmable FSR of ±2g, ±4g, ±8g, and ±16g
- 3-Axis Compass with a wide range to ±4900 µT
- Onboard Digital Motion Processor (DMP)
- Android support
- Auxiliary I^2C interface for external sensors
- On-Chip 16-bit ADCs and Programmable Filters
- 7 MHz SPI or 400 kHz Fast Mode I^2C
- Digital-output temperature sensor
- VDD operating range of 1.71V to 3.6V
- MEMS structure hermetically sealed and bonded at wafer level
- RoHS and Green compliant

TYPICAL OPERATING CIRCUIT



TDK InvenSense

ICM-20948

3 ELECTRICAL CHARACTERISTICS

3.1 GYROSCOPE SPECIFICATIONS

Typical Operating Circuit of section 4.2, VDD = 1.8V, VDDIO = 1.8V, T_A=25°C, unless otherwise noted.

NOTE: All specifications apply to Low-Power Mode and Low-Noise Mode, unless noted otherwise

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
GYROSCOPE SENSITIVITY						
Full-Scale Range	GYRO_FS_SEL=0		±250		dps	1
	GYRO_FS_SEL=1		±500		dps	1
	GYRO_FS_SEL=2		±1000		dps	1
	GYRO_FS_SEL=3		±2000		dps	1
Gyroscope ADC Word Length			16		bits	1
Sensitivity Scale Factor	GYRO_FS_SEL=0		131		LSB/(dps)	1
	GYRO_FS_SEL=1		65.5		LSB/(dps)	1
	GYRO_FS_SEL=2		32.8		LSB/(dps)	1
	GYRO_FS_SEL=3		16.4		LSB/(dps)	1
Sensitivity Scale Factor Tolerance	25°C		±1.5		%	2
Sensitivity Scale Factor Variation Over Temperature	-40°C to +85°C		±3		%	2
Nonlinearity	Best fit straight line; 25°C		±0.1		%	2, 3
Cross-Axis Sensitivity			±2		%	2, 3
ZERO-RATE OUTPUT (ZRO)						
Initial ZRO Tolerance	25°C (Component-level)		±5		dps	2
ZRO Variation Over Temperature	-40°C to +85°C		±0.05		dps/°C	2
GYROSCOPE NOISE PERFORMANCE (GYRO_FS_SEL=0)						
Noise Spectral Density	Based on Noise Bandwidth = 10 Hz		0.015		dps/√Hz	2
GYROSCOPE MECHANICAL FREQUENCIES		25	27	29	kHz	2
LOW PASS FILTER RESPONSE	Programmable Range	5.7		197	Hz	1, 3
GYROSCOPE START-UP TIME	From Full-Chip Sleep mode		35		ms	2, 3
OUTPUT DATA RATE	Low-Power Mode	4.4		562.5	Hz	1
	Low-Noise Mode GYRO_FCHOICE=1; GYRO_DLPFCFG=x	4.4		1.125k	Hz	
	Low-Noise Mode GYRO_FCHOICE=0; GYRO_DLPFCFG=x			9k	Hz	

Table 1. Gyroscope Specifications

NOTES:

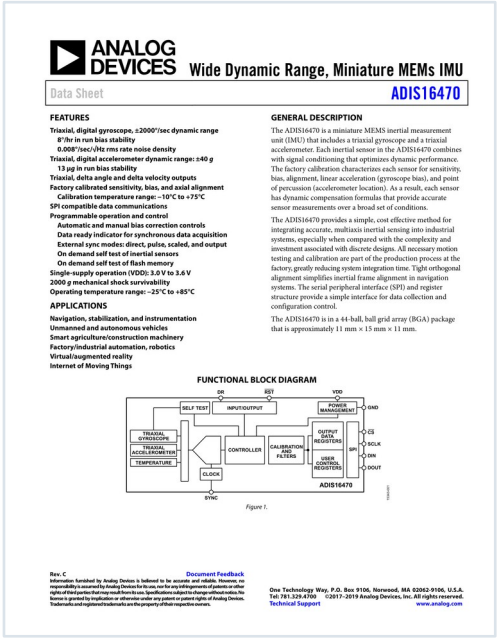
1. Guaranteed by design.
2. Derived from validation or characterization of parts, not guaranteed in production.
3. Low-noise mode specification.

TDK InvenSense, ICM-20948 Data Sheet, DS-000189 v1.3 (2017), p.1 與 p.11 · 原頁全文 · 讀取 2026/10/08 · 版權歸 TDK ，課堂引用

Phụ lục · 附錄（不計時）

Analog Devices ADIS16470 — trang bìa và trang 3 (Table 1)

Analog Devices ADIS16470 — 封面與第 3 頁（Table 1）



ANALOG DEVICES

Wide Dynamic Range, Miniature MEMs IMU

Data Sheet ADIS16470

FEATURES

- Triaxial, digital gyroscope, ±2000°/sec dynamic range
  - 8°/hr in run bias stability
  - 0.008°/sec/√Hz rms rate noise density
- Triaxial, digital accelerometer dynamic range: ±40 g
  - 13 µg in run bias stability
- Triaxial, delta angle and delta velocity outputs
- Factory calibrated sensitivity, bias, and axial alignment
  - Calibration temperature range: −10°C to +75°C
- SPI compatible data communications
- Programmable operation and control
  - Automatic and manual bias correction controls
  - Data ready indicator for synchronous data acquisition
  - External sync modes: direct, pulse, scaled, and output
  - On demand self test of inertial sensors
  - On demand self test of flash memory
- Single-supply operation (VDD): 3.0 V to 3.6 V
- 2000 g mechanical shock survivability
- Operating temperature range: −25°C to +85°C

APPLICATIONS

- Navigation, stabilization, and instrumentation
- Unmanned and autonomous vehicles
- Smart agriculture/construction machinery
- Factory/industrial automation, robotics
- Virtual/augmented reality
- Internet of Moving Things

GENERAL DESCRIPTION

The ADIS16470 is a miniature MEMS inertial measurement unit (IMU) that includes a triaxial gyroscope and a triaxial accelerometer. Each inertial sensor in the ADIS16470 combines with signal conditioning that optimizes dynamic performance. The factory calibration characterizes each sensor for sensitivity, bias, alignment, linear acceleration (gyroscope bias), and point of percussion (accelerometer location). As a result, each sensor has dynamic compensation formulas that provide accurate sensor measurements over a broad set of conditions.

The ADIS16470 provides a simple, cost effective method for integrating accurate, multiaxis inertial sensing into industrial systems, especially when compared with the complexity and investment associated with discrete designs. All necessary motion testing and calibration are part of the production process at the factory, greatly reducing system integration time. Tight orthogonal alignment simplifies inertial frame alignment in navigation systems. The serial peripheral interface (SPI) and register structure provide a simple interface for data collection and configuration control.

The ADIS16470 is in a 44-ball, ball grid array (BGA) package that is approximately 11 mm × 15 mm × 11 mm.

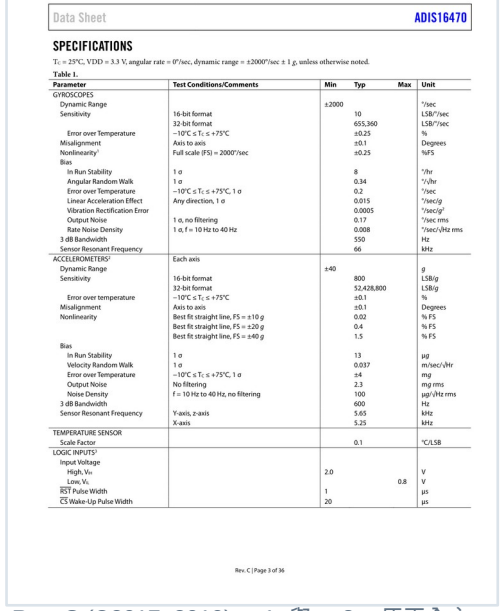
FUNCTIONAL BLOCK DIAGRAM

Figure 1.

Rev. C Document Feedback

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SPECIFICATIONS

T_C = 25°C, VDD = 3.3 V, angular rate = 0°/sec, dynamic range = ±2000°/sec ± 1 g, unless otherwise noted.

Table 1.

Parameter	Test Conditions/Comments	Min	Typ	Max	Unit
GYROSCOPES					
Dynamic Range		±2000			°/sec
Sensitivity	16-bit format		10		LSB/°/sec
	32-bit format		655,360		LSB/°/sec
Error over Temperature	−10°C ≤ T_C ≤ +75°C		±0.25		%
Misalignment	Axis to axis		±0.1		Degrees
Nonlinearity[1]	Full scale (FS) = 2000°/sec		±0.25		%FS
Bias					
In Run Stability	1 σ		8		°/hr
Angular Random Walk	1 σ		0.34		°/√hr
Error over Temperature	−10°C ≤ T_C ≤ +75°C, 1 σ		0.2		°/sec
Linear Acceleration Effect	Any direction, 1 σ		0.015		°/sec/g
Vibration Rectification Error			0.0005		°/sec/g²
Output Noise	1 σ, no filtering		0.17		°/sec rms
Rate Noise Density	1 σ, f = 10 Hz to 40 Hz		0.008		°/sec/√Hz rms
3 dB Bandwidth			550		Hz
Sensor Resonant Frequency			66		kHz
ACCELEROMETERS[2]	Each axis				
Dynamic Range		±40			g
Sensitivity	16-bit format		800		LSB/g
	32-bit format		52,428,800		LSB/g
Error over temperature	−10°C ≤ T_C ≤ +75°C		±0.1		%
Misalignment	Axis to axis		±0.1		Degrees
Nonlinearity	Best fit straight line, FS = ±10 g		0.02		% FS
	Best fit straight line, FS = ±20 g		0.4		% FS
	Best fit straight line, FS = ±40 g		1.5		% FS
Bias					
In Run Stability	1 σ		13		µg
Velocity Random Walk	1 σ		0.037		m/sec/√Hr
Error over Temperature	−10°C ≤ T_C ≤ +75°C, 1 σ		±4		mg
Output Noise	No filtering		2.3		mg rms
Noise Density	f = 10 Hz to 40 Hz, no filtering		100		µg/√Hz rms
3 dB Bandwidth			600		Hz
Sensor Resonant Frequency	Y-axis, z-axis		5.65		kHz
	X-axis		5.25		kHz
TEMPERATURE SENSOR					
Scale Factor			0.1		°C/LSB
LOGIC INPUTS[3]					
Input Voltage					
High, V_{IH}		2.0			V
Low, V_{IL}				0.8	V
$\overline{RST}$ Pulse Width		1			µs
$\overline{CS}$ Wake-Up Pulse Width		20			µs

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